

Sec 3.4 – The Chain Rule

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

Imagine we are moving straight upward in a hot air balloon. Let y be our distance from the ground. The temperature, H , is changing as a function of altitude, so $H = f(y)$.

Suppose we are moving upward at a rate of 1.5 miles an hour and the temperature is changing at a rate of 16 degrees Fahrenheit per mile of altitude. How is our temperature changing with time?

Since temperature is a function of height, $H = f(y)$, and height is a function of time, $y = g(t)$, we can think of temperature as a composite function of time, $H = f(g(t))$, with f as the outside function and g as the inside function.

Recall what function composition is:

$$f \circ g(x) = f(g(x)),$$

which means that the output of the function $g(x)$ becomes the input of the function $f(x)$.

Find $f(g(x))$ and $g(f(x))$ for each of the following pairs of functions:

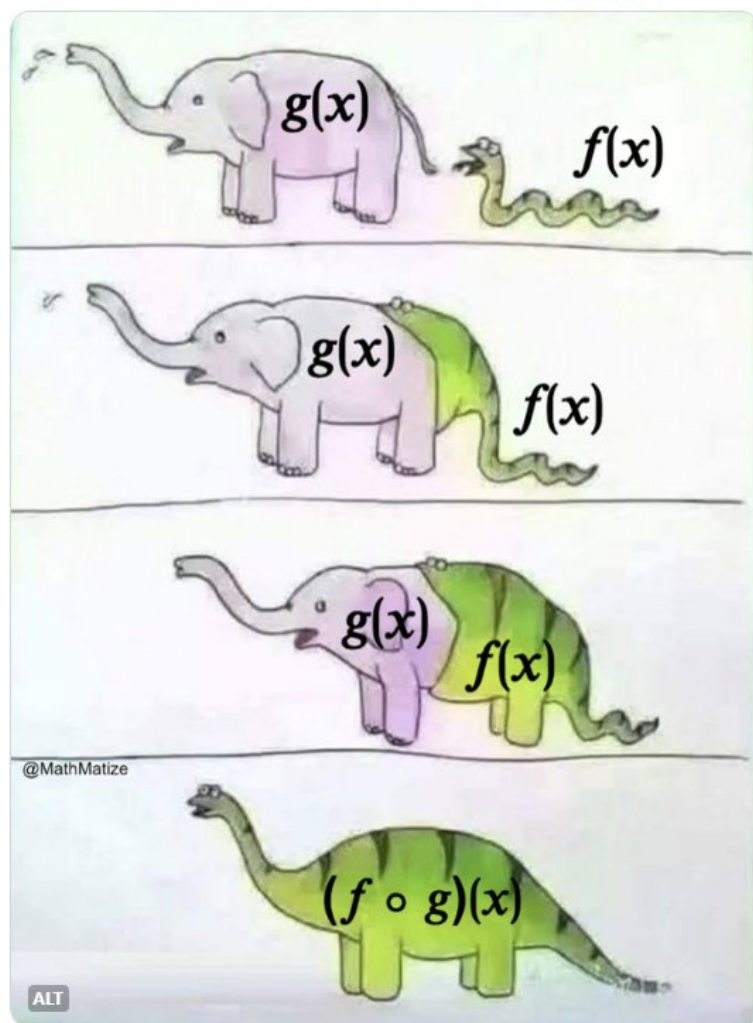
1. $f(x) = x^4$, $g(x) = x + 3$

$$f(g(x)) = (x + 3)^4 \quad g(f(x)) = x^4 + 3$$

2. $f(x) = e^x$, $g(x) = x^2 + 3x$

$$f(g(x)) = e^{x^2+3x}, \quad g(f(x)) = (e^x)^2 + 3(e^x) = e^{2x} + 3e^x$$

Dinosaurs are a composition of snakes and elephants 🐍🐘🦖



9:17 AM · Oct 27, 2023 · 22.9K Views

Write the following as a composition of functions:

1. $h(x) = 3^{x^2+5} + 10$:

Let $f(x) = 3^x + 10$ and $g(x) = x^2 + 5$. Then $h(x) = f(g(x))$.

2. $h(x) = \frac{1}{x^3 + 3x^6}$:

Let $f(x) = x^3 + 3x^6$ and $g(x) = \frac{1}{x}$. Then $h(x) = g(f(x))$.

Note that $g(x) = \frac{1}{x} = x^{-1}$.

So, back to our original example,

Our example with the hot air balloon suggested that

$$\begin{array}{ccccc} \text{Rate of change} & = & \text{Rate of change} & \times & \text{Rate of change} \\ \text{of composite function} & & \text{of outside function} & & \text{of inside function} \end{array}$$

To write this in terms of function notation,

Theorem (The Chain Rule)

$$\frac{dy}{dx} = \frac{dy}{dz} \frac{dz}{dx} \quad \text{or} \quad \frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$$

In words: the derivative of a composite function is the product of the derivatives of the outside and inside functions.

Note: the derivative of the outside function must be evaluated at the inside function.

Use the chain rule to find the following derivatives:

$$h(x) = \sqrt{x^2 + 1} = (x^2 + 1)^{1/2}$$

$$h(x) = e^{x^2}$$

$$h'(x) = \sqrt{x^2 + 1} = \frac{1}{2}(x^2 + 1)^{-1/2} \cdot 2x$$

$$h'(x) = e^{x^2} \cdot 2x$$

Partner up

Find the derivatives:

1. $h(x) = (x + 3)^4$

2. $h(x) = 3^{x^2+5} + 10$

3. $h(x) = (x^3 - 3x^2 + 5)^{-1}$

4. $h(x) = \sqrt[3]{e^{x^2+2x} + x^3}$

5. $h(x) = 5^{\sqrt{x^3+1}}$

6. $h(x) = (x^2 + 5)e^{3x}$

Suppose $h(x) = (g(x))^3$. You know that $g(3) = 5$ and $g'(3) = -4$. What is $h'(3)$?

$$1. h(x) = (x + 3)^4 \quad h'(x) = 4(x + 3)^3(x + 3)' = 4(x + 3)^3(1)$$

$$2. h(x) = 3^{x^2+5} + 10$$

$$h'(x) = \left((\ln 3)3^{x^2+5} \right) (x^2 + 5)' = \left((\ln 3)3^{x^2+5} \right) (2x)$$

$$3. h(x) = (x^3 - 3x^2 + 5)^{-1}$$

$$h'(x) = -1(x^3 - 3x^2 + 5)^{-2}(x^3 - 3x^2 + 5)'$$

$$= -1(x^3 - 3x^2 + 5)^{-2}(3x^2 - 6x)$$

$$4. h(x) = \sqrt[3]{e^{x^2+2x} + x^3} = (e^{x^2+2x} + x^3)^{1/3}$$

We use the chain rule multiple times:

$$h'(x) = (1/3)(e^{x^2+2x} + x^3)^{-2/3}(e^{x^2+2x} + x^3)'$$

$$= (1/3)(e^{x^2+2x} + x^3)^{-2/3}(e^{x^2+2x}(x^2 + 2x)' + 3x^2)$$

$$= (1/3)(e^{x^2+2x} + x^3)^{-2/3}(e^{x^2+2x}(2x + 2) + 3x^2)$$

$$5. h(x) = 5^{\sqrt{x^3+1}}$$

$$h'(x) = (\ln 5)5^{\sqrt{x^3+1}}(\sqrt{x^3+1})'$$

$$= (\ln 5)5^{\sqrt{x^3+1}}((1/2)(x^3 + 1)^{-1/2})(x^3 + 1)'$$

$$= (\ln 5)5^{\sqrt{x^3+1}}((1/2)(x^3 + 1)^{-1/2})(3x^2)$$

$$6. h(x) = (x^2 + 5)e^{3x} \text{ We use the product rule first:}$$

$$h'(x) = (x^2 + 5)'e^{3x} + (x^2 + 5)(e^{3x})' = 2xe^{3x} + (x^2 + 5)(e^{3x}(3x)')$$

$$= 2xe^{3x} + (x^2 + 5)(e^{3x}(3))$$

Suppose $h(x) = (g(x))^3$. You know that $g(3) = 5$ and $g'(3) = -4$. What is $h'(3)$?

$$h'(x) = 3(g(x))^2 g'(x). \text{ Therefore}$$

$$h'(3) = 3(g(3))^2 g'(3) = 3(5)^2(-4) = -300$$